Field Study Report: Gaziantep Metropolitan Municipality Website

Site	https://www.gaziantep.bel.tr/
Organisation	Gaziantep Büyükşehir Belediyesi (Gaziantep Metropolitan Municipality)
Student	Hayrunnisa İYİKÖŞKER, 2203060011
Course	SFWE475 Advanced Web Programming, Assignment 2 (Core tier)
Audit date	7–8 November 2026
Browser	Google Chrome 156.0.8078.80 (64-bit), Windows 11 (Lighthouse run in Mobile mode)

Part A: Under the hood

A1. HTTPS and redirects. When I typed the address with http://, the server answered with 301 Moved Permanently and sent my browser to https://www.gaziantep.bel.tr/. This page then loaded normally with status 200. The padlock shows a secure connection and a valid certificate. It was issued by Sectigo to *.gaziantep.bel.tr (a wildcard certificate, so it also covers subdomains). It is valid from 19 February 2026 and expires on 20 February 2027. So visitors who use an old http:// link still reach the secure site, and the certificate is fine now. It expires in about three and a half months, so the municipality should renew it on time. [A1]

A2. URL structure. I scored five pages against the five URL rules from Week 5.

#	URL	HTTPS	Short and readable	Lowercase	Hyphens	Few parameters	Overall
1	https://www.gaziantep.bel.tr/ (home page)	Pass	Pass	Pass	N/A (no words)	Pass (0)	Pass
2	https://www.gaziantep.bel.tr/tr/basin/duyurular	Pass	Pass	Pass	N/A (single words)	Pass (0)	Pass
3	https://gaziantep.bel.tr/tr/projeler	Pass	Pass	Pass	N/A	Pass (0)	Pass
4	https://gaziantep.bel.tr/tr/projeler/abdulhamit-han-koprulu-kavsagiyla-trafikte-yeni-donem	Pass	Partial (slug is long)	Pass	Pass (words joined with -, no _)	Pass (0)	Partial
5	https://www.gaziantep.bel.tr/tr/basin/etkinlikler?name=tiyatro	Pass	Pass	Pass	N/A	Pass (1: name)	Pass

Pass: the URL meets the rule. Partial: it mostly meets the rule but could be better. N/A: the rule does not apply.

All five addresses use HTTPS, are lowercase, and have no or very few query parameters. The project page has the most readable address. Its slug uses hyphens and describes the page title, and Turkish letters appear as plain letters (for example “koprulu”), so there are no unreadable codes. The only parameter I found was ?name=tiyatro on the events filter. It is short and easy to understand. The one weak point is the long project slug (53 characters), which is still readable. I also saw that some pages open on www.gaziantep.bel.tr and others on gaziantep.bel.tr, so the site seems to use two forms of its main address. I checked which one it prefers in A7. [A2]

A3. One request, closely. I reopened the home page with the Network panel open and the cache disabled. This is like a first visit. The first request was the HTML document. It used the GET method and returned status 200, and the content type was text/html; charset=UTF-8. The HTML file was small (about 76 kB), but the whole page made 52 requests and transferred 12.9 MB (14.5 MB in total resources). DOMContentLoaded fired at 2.87 seconds and the full page loaded at 6.35 seconds. The only Fetch/XHR requests I found were POST requests to Google Analytics. They returned status 204 (No Content), so they did not carry page content. The page is quite heavy. The three biggest files were two parts of a video (gaziantep.webm, 4.2 MB and 2.4 MB) and one GIF (chatbot2.gif, 2.3 MB), which is used for the chat button. Together they are about 8.9 MB, about two-thirds of the transferred data. This can make the site slow on mobile devices with a slow connection. I look at this more in B1. [A3]

A4. A page that does not exist. I opened a made-up address: /no-such-page-123. The server returned status 404 (Not Found), which is correct because the page does not exist. The visitor sees an error page with the municipality logo and the title “Page Not Found!”, and a short message that the page may have

been removed or moved. The page gives useful options, such as the buttons “Go to homepage” and “Go back”. It also has Quick Links, contact cards and a bottom navigation bar. There is one problem: the message is only in English. I tried to change the language to Turkish with the bottom menu, but the page stayed in English. This may confuse Turkish visitors on a Turkish municipality site. [A4]

A5. Caching. The main page uses Cache-Control: private, must-revalidate. This means the browser may keep the page on the visitor’s device, but it must check with the server that the saved copy is still up to date before using it. For the image a314e3fd-...original.JPG (served from dosya.gaziantep.bel.tr), the server response had no Cache-Control header. The no-cache line I saw in the request headers was added by my browser, because I had disabled the cache. The server did send ETag and Last-Modified headers, so the browser can check if the image changed. But there is no clear rule for how long the browser may keep the image. This is a large image that probably does not change often. A longer cache time (for example Cache-Control: public, max-age=...) would help, because visitors would not need to download it again on later visits. [A5]

A6. Where the page is built. I opened the page source with Ctrl+U and searched for text that I can see on the page. The menu word “haberler” and the news title “GAZİANTEP BÜYÜKŞEHİR’DEN ULAŞIM HİZMETLERİNDE YENİ DİJİTAL HAMLE” were already in the HTML source. The Elements panel showed the same text (4 matches for the news title). So the server sends the main content as ready HTML, and I think the home page is server-rendered. The page still loads many JavaScript files, probably for things like the news sliders, but the main text is there without JavaScript. This helps visitors on slow connections and search engines, which can read the content without running JavaScript. [A6]

A7. SEO basics. I checked the home page for the basic SEO items. The page has a clear title (“Gaziantep Büyükşehir Belediyesi”) and a meta description that explains what the website is for. It has a canonical link that points to https://www.gaziantep.bel.tr/. The page language is Turkish (lang="tr") and there is only one h1 heading. I did not find any rel="sponsored" or rel="ugc" links. This is expected, because the site does not seem to have adverts or user comments. All the basic SEO items are present, and I did not find a big SEO problem on the home page. The canonical link on the home page points to the www address, so the site seems to prefer it. This fits what I saw in A2, where some pages opened without www. [A7]

Part B: The experience

B1. Loading state. I reloaded the home page with the cache disabled and chose Chrome’s slowest throttling preset, named “3G” (in this Chrome version it is the lowest preset, and the old “Slow 3G” name is not shown). For about 13 seconds I saw only a loading screen with the municipality logo and three animated dots. The page was not blank, but I could not use anything. After that, the text and layout appeared, but the images loaded much slower. After one minute only about 3.3 MB was downloaded and many news images were still loading. One image was only partly visible, and there was no placeholder while it loaded. The small icons showed as empty boxes, probably because the icon font was not loaded yet. I could not see the background video. This can be a problem for visitors with slow internet: they wait a long time, and after the page appears some parts are still missing. [B1]

B2. Empty state. On the events page, I typed “zzzqqq” in the search box and clicked the search button. The page did not show an error or a blank area. It showed an icon and the message “Kayıt bulunamadı” (“No records found”), and a button “Tüm Etkinlikler” (“All Events”). The filter part showed “Toplam: 0 kayıt” (“Total: 0 records”) and a “Temizle” (“Clear”) button. The text I typed stayed in the search box, so visitors can correct their search easily. I think this is a clear and friendly empty state, because it gives the visitor another option. It could be better if the message also said to check the spelling or try another word. Also, “Sayfa: 1 / 1” (“Page: 1 / 1”) looks a little odd when there are no results. This message is in Turkish, but the 404 page in A4 was in English, so the language is not consistent across the site. [B2]

B3. Error state. I set the network to Offline in DevTools (with the cache disabled) and then opened another page of the site, /tr/basin/duyurular?name=tapu. The visitor did not see any message from the website. Chrome showed its own error page (ERR_INTERNET_DISCONNECTED) with the offline dinosaur game (“Press space to play”). The page also has a “Reload” button. This is expected when a new page cannot be loaded at all, because the website’s own code cannot run without a connection. So I cannot say that the site handles this badly. I did not check if the site has an offline page. After the test I set the network back to online. [B3]

B4. Form validation. I opened the “Tesis Ziyaret Formu” (“Facility Visit Form”). Before I clicked submit, I could not tell which fields were required, because there were no asterisks, red marks or “required” labels. I typed only one letter in the first field (“Ziyarette Bulunacak Kurum / Kişi”) and left the “Adı” field empty. When I clicked submit, a small message appeared next to the empty field: “Please fill out this field.” This message is in English and seems to come from Chrome’s built-in validation, not from the website. Turkish visitors may see it in Turkish, depending on their browser language. Showing the error next to the empty field is good, and the text I typed in the first field was not deleted. But visitors cannot see which fields are required before they submit. The message also shows only one missing field at a time and does not say what to enter. [B4]

B5. Labels. On the “Tesis Ziyaret Formu”, every field has a visible label above it, such as “Adı”, “Soyadı”, “E-Posta” and “Telefon”. The phone field also shows an example format as a placeholder, which is helpful. But when I clicked the label “Adı”, nothing happened: the cursor did not move into the input field. The field became active only when I clicked inside the input box. The other fields I tested did the same. I checked the Elements panel and found the reason. The <label> for “Adı” has no for attribute and the <input> has no id, so they are not connected in the page code. The input does have a required attribute, which fits my finding in B4: the field is required, but the website does not show this to visitors. Labels like this are less useful for mouse users, who cannot click a label to focus its field. They can also cause problems for screen reader users, because the field name may not be read out correctly. [B5]

B6. Keyboard only. My task was to find the organisation chart (“Teşkilat Şeması”) using only the Tab, Shift+Tab and Enter keys. The focus outline was easy to see and the link order was logical, from left to right. But when I pressed Tab for the first time, I could not find a “skip to content” link. When I kept pressing Tab, the focus went into the news slider and got stuck. Even after about 25 presses I could not leave the slider with Tab. I had to press Shift+Tab to go back, and then press Enter on a link to open the page. This looks like a keyboard trap. It can be a serious problem for visitors who cannot use a mouse, such as people with motor disabilities, and for screen reader users, who may get stuck in the slider and not reach the rest of the page. A “skip to content” link would also help keyboard users skip the menu and the slider. [B6]

B7. Automated audit. I used Lighthouse to check the accessibility of the home page and the events page in Mobile mode. The home page scored 68/100 and the events page scored 67/100. Both pages need improvement.

Main problems on the home page:

- Buttons without accessible names
- Images without alt text
- Form fields without labels
- Links without discernible names

Main problems on the events page:

- Form fields without labels
- user-scalable="no" in the viewport settings, which can stop users from zooming on their phones
- No main landmark on the page
- Some text with not enough contrast against its background

The missing form labels on both pages fit what I found in my manual test in B5. The missing main landmark is also related to my keyboard test in B6, where I could not find a “skip to content” link. I only tested in Mobile mode, so the scores may be different on desktop. Lighthouse also cannot find every problem. For example, it did not find the keyboard trap in the news slider, so I also did manual tests. [B7]

B8. Link and button text. Lighthouse found 10 buttons without accessible names on the home page. I checked them and they are the round eye-icon buttons at the bottom right of each card in the “Duyurular” (“Announcements”) slider. In the page code, these buttons have only an icon, with no text or label. Screen reader users may hear only the word “button” and not know what it does. The arrow buttons in the slider do have names, but they are in English (“Previous slide” and “Next slide”). The site is in Turkish, so Turkish names would be better. I also checked the chat robot button at the bottom right of the page. It uses only a

background image (the same chatbot2.gif from A3), and I could not find any text or label for it. The table shows the elements I checked and the names I suggest. [B8]

#	Element and location	Current text	Better text
1	Home page, bottom right of each announcement card (10 buttons)	None (eye icon only)	“Duyuruyu oku” (“Read the announcement”)
2	Announcements slider, left arrow	“Previous slide”	“Önceki duyuru” (“Previous announcement”)
3	Announcements slider, right arrow	“Next slide”	“Sonraki duyuru” (“Next announcement”)
4	Chat robot button, fixed at the bottom right	None (background image only)	“Canlı sohbeti aç” (“Open live chat”)

Part C: Recommendations

What works well

1. The website sends old http:// links to https:// with a 301 status, and it has a valid certificate, so visitors use the secure version of the site [A1].
2. The empty search result is clear and friendly. When I searched for “zzzqqq”, the page showed a message and a “Tüm Etkinlikler” (“All Events”) button, and my text stayed in the box, so I could correct it easily [B2].

Three problems to fix first

I put the keyboard problem first because some visitors cannot continue using the website. The heavy home page is second because it makes the site slow, especially on slow connections. The buttons and form fields without names or labels are third because they are easier to fix.

1. The news slider traps keyboard users

- **Problem:** When I used the Tab key, the focus got stuck in the news slider and I could not move on to the rest of the page.
- **Evidence:** [B6], [B7].
- **Who it affects:** People who cannot use a mouse, including people with motor disabilities and screen reader users.
- **Fix:** Make it possible to move past the slider with the Tab key, and add a “skip to content” link at the top of the page.
- **Effort and impact:** Effort: medium. Impact: high.

2. The home page is too heavy

- **Problem:** The home page makes 52 requests and downloads about 12.9 MB. Two parts of a video and one GIF are about two-thirds of this size. On a slow connection I waited about 13 seconds for the page to become usable, and many images were still loading.
- **Evidence:** [A3], [B1], [A5].
- **Who it affects:** Visitors who use mobile data or a slow internet connection.
- **Fix:** Compress the large GIF and the video, do not load the background video on mobile phones, load images only when they are needed (lazy loading), and give images a longer cache time.
- **Effort and impact:** Effort: medium. Impact: high.

3. Buttons and form fields have no names or labels

- **Problem:** The eye-icon buttons and the chat button have no accessible names. In the form, the labels are not connected to their fields, and required fields are not shown before submitting.
- **Evidence:** [B4], [B5], [B7], [B8].
- **Who it affects:** Screen reader users and people who use the forms.
- **Fix:** Give a clear name to every icon button, connect each label to its input field, and mark required fields with a star (*) and a short note.
- **Effort and impact:** Effort: low. Impact: medium.

Other finding

The 404 page and its language menu stay in English even though this is a Turkish website [A4]. Other parts of the site show messages in Turkish [B2].

Appendix A: Evidence log

One row for each piece of evidence. The files are in the evidence/ folder of the repository. Times are from my computer clock (dd.mm.yyyy). Rows that start with S1 are the screenshots of the second site for Stretch S1 (Şahinbey Belediyesi).

ID	File	What it shows	Time and date	Page address
A1	evidence/A1-redirect.png	http:// request answered with 301 redirect to https://	07.11.2026 16:49	http://www.gaziantep.bel.tr/
A1	evidence/A1-padlock.png	Padlock: connection secure, certificate valid	07.11.2026 16:59	https://www.gaziantep.bel.tr/
A1	evidence/A1-certificate.png	Certificate issuer and validity dates	07.11.2026 16:59	https://www.gaziantep.bel.tr/
A2	evidence/A2-url1.png	Home page address	07.11.2026 17:22	https://www.gaziantep.bel.tr/
A2	evidence/A2-url2.png	Announcements page address	07.11.2026 17:23	https://www.gaziantep.bel.tr/tr/basin/duyurular
A2	evidence/A2-url3.png	Projects list address (no www)	07.11.2026 17:23	https://gaziantep.bel.tr/tr/projeler
A2	evidence/A2-url4.png	Project detail address with long slug	07.11.2026 17:24	https://gaziantep.bel.tr/tr/projeler/abdulhamit-han-koprulu-kavsagiyla-trafikte-yeni-donem
A2	evidence/A2-url5.png	Events page with one query parameter	07.11.2026 17:25	https://www.gaziantep.bel.tr/tr/basin/etkinlikler?name=tiyatro
A3	evidence/A3-headers.png	Headers panel of the first (document) request	07.11.2026 20:15	https://www.gaziantep.bel.tr/
A3	evidence/A3-summary.png	Network summary bar: number of requests and total size (cache disabled)	07.11.2026 20:22	https://www.gaziantep.bel.tr/
A3	evidence/A3-xhr.png	One Fetch/XHR request and its headers	07.11.2026 21:14	https://www.gaziantep.bel.tr/
A3	evidence/A3-largest.png	Network list sorted by size: largest files are a video and a GIF	07.11.2026 21:36	https://www.gaziantep.bel.tr/
A4	evidence/A4-missingpage.png	404 page: message and homepage button; Network shows status 404 for the document	07.11.2026 21:40	https://www.gaziantep.bel.tr/no-such-page-123
A4	evidence/A4-language.png	404 page still in English after choosing Turkish in the language menu	07.11.2026 21:46	https://www.gaziantep.bel.tr/no-such-page-123
A5	evidence/A5-cache.png	Response headers of the main document: Cache-Control private, must-revalidate	07.11.2026 22:03	https://www.gaziantep.bel.tr/
A5	evidence/A5-cache-image.png	Response headers of one image: no Cache-Control; ETag and Last-Modified present	07.11.2026 22:05	https://dosya.gaziantep.bel.tr/uploads/new/2026/10/a314e3fd-de6a-4278-9f8e-640659023ed0_original.JPG
A6	evidence/A6-source.png	Raw page source (Ctrl+U): the menu word “haberler” is found	07.11.2026 23:10	view-source:https://www.gaziantep.bel.tr/
A6	evidence/A6-elements.png	Elements panel: same menu word found	07.11.2026 23:11	https://www.gaziantep.bel.tr/
A6	evidence/A6-source-news.png	Raw page source: a news headline from the slider is found	07.11.2026 23:13	view-source:https://www.gaziantep.bel.tr/
A6	evidence/A6-elements-news.png	Elements panel: same headline found	07.11.2026 23:14	https://www.gaziantep.bel.tr/
A7	evidence/A7-seo.png	Console checks: title, meta description, canonical, lang, h1 count, sponsored/ugc count	08.11.2026 01:18	https://www.gaziantep.bel.tr/
B1	evidence/B1-loading-1.png	Loading screen (logo and three dots), about 11 s after reload with the 3G preset	08.11.2026 01:27	https://www.gaziantep.bel.tr/
B1	evidence/B1-loading-2.png	Page content visible, icons shown as empty boxes, about 30 s after reload	08.11.2026 01:27	https://www.gaziantep.bel.tr/
B1	evidence/B1-loading-3.png	News photo only partly drawn, many images still pending, about 1 min after reload	08.11.2026 01:28	https://www.gaziantep.bel.tr/
B2	evidence/B2-empty.png	Events filter with “zzzqqq”: message “Kayıt bulunamadı”, “Tüm Etkinlikler” button, total 0 records	08.11.2026 14:45	https://www.gaziantep.bel.tr/tr/basin/etkinlikler?name=zzzqqq
B3	evidence/B3-offline.png	Chrome’s own offline error page shown when opening another page of the site while Offline (cache disabled)	08.11.2026 16:55	https://www.gaziantep.bel.tr/tr/basin/duyurular?name=tapu

ID	File	What it shows	Time and date	Page address
B4	evidence/B4-form-errors.png	Visit form submitted with “Adı” empty: browser message “Please fill out this field.” under the field; no required marks before submit	08.11.2026 17:46	https://www.gaziantep.bel.tr/tr/tesis-ziyaret-formu
B5	evidence/B5-labels.png	Visit form: clicking the label “Adı” does not move focus into the field	08.11.2026 17:59	https://gaziantep.bel.tr/tr/tesis-ziyaret-formu
B5	evidence/B5-label-code.png	Elements panel: <label> for “Adı” has no for attribute; <input> has no id but has required	08.11.2026 18:00	https://gaziantep.bel.tr/tr/tesis-ziyaret-formu
B6	evidence/B6-focus-visible.png	Keyboard only: focus outline clearly visible on a menu link	08.11.2026 22:18	https://www.gaziantep.bel.tr/
B6	evidence/B6-first-tab.png	First Tab press: no “skip to content” link appears	08.11.2026 22:18	https://www.gaziantep.bel.tr/
B6	evidence/B6-trap-1.png	Focus inside the news slider (keyboard trap), first moment	08.11.2026 22:19	https://www.gaziantep.bel.tr/
B6	evidence/B6-trap-2.png	Focus still inside the news slider after more Tab presses	08.11.2026 22:20	https://www.gaziantep.bel.tr/
B6	evidence/B6-result.png	Organisation chart page reached with Shift+Tab then Enter	08.11.2026 22:20	https://www.gaziantep.bel.tr/tr/teskilat-semasi
B6	evidence/B6-keyboard-log.txt	Short log: task, keys, problems	08.11.2026 22:21	https://www.gaziantep.bel.tr/
B7	evidence/B7-lighthouse-home.png	Lighthouse Accessibility report for the home page: score 68, issues listed	08.11.2026 22:45	https://www.gaziantep.bel.tr/
B7	evidence/B7-lighthouse-inner.png	Lighthouse Accessibility report for the events page: score 67, issues listed	08.11.2026 22:58	https://www.gaziantep.bel.tr/tr/basin/etkinlikler?name=tiyatro
B8	evidence/B8-lighthouse-buttons.png	Lighthouse lists 10 buttons without an accessible name (same class “btn-circle”)	08.11.2026 22:45	https://www.gaziantep.bel.tr/
B8	evidence/B8-eye-button-code.png	Elements: announcement card button has only an eye icon, no text or aria-label	08.11.2026 23:35	https://www.gaziantep.bel.tr/
B8	evidence/B8-arrows-code.png	Elements: slider arrow buttons have aria-label “Previous slide” / “Next slide” in English	08.11.2026 23:40	https://www.gaziantep.bel.tr/
B8	evidence/B8-chatbot-code.png	Elements: chat robot button uses a background image (chatbot2.gif), no visible text or aria-label	08.11.2026 23:46	https://www.gaziantep.bel.tr/
S1-A1	evidence/S1-A1-site2.png	Second site: home page open on https://www.sahinbey.bel.tr/Default (status of the http:// request not recorded)	09.11.2026 01:20	https://www.sahinbey.bel.tr/Default
S1-A3	evidence/S1-A3-site2.png	Second site: Network summary bar, 152 requests, 30.3 MB transferred (cache disabled)	09.11.2026 01:22	https://www.sahinbey.bel.tr/Default
S1-A4	evidence/S1-A4-site2.png	Second site: made-up address answered with status 200 and a plain text page	09.11.2026 01:25	https://www.sahinbey.bel.tr/sayfayok?aspxerrorpath=/Default/no-such-page-123.aspx
S1-A7	evidence/S1-A7-site2.png	Second site: console checks for the SEO basics	09.11.2026 01:27	https://www.sahinbey.bel.tr/Default
S1-B2	evidence/S1-B2-site2.png	Second site: search for “zzzqqq” in the map application: “Kayıt bulunamadı.”	09.11.2026 01:28	https://kbs.sahinbey.bel.tr/keos/
S1-B7	evidence/S1-B7-site2.png	Second site: Lighthouse Accessibility score 69, with a warning that results may be incomplete	09.11.2026 01:16	https://www.sahinbey.bel.tr/Default